Predicting Machine Failure

DATA DIVERGENCE · Track 1 (Beginner) | Team Gradient Descenders | Dharunesh A & Valliappan S | Methodology summary

0.931	0.914	99.6%	84.5%
PR-AUC ranking quality	F1 at the 0.5 cut-off	precision at most 1 false alarm	recall 229 of 271 caught

A logistic regression on five features built from the physics of the machine. Scored only on rows it had not seen while training, it found **229 of the 271 failures** and raised **at most one false alarm**. The 42 it missed are 35 tool-wear failures that only happen some of the time and 7 with no visible cause, which is close to the most this data can give.

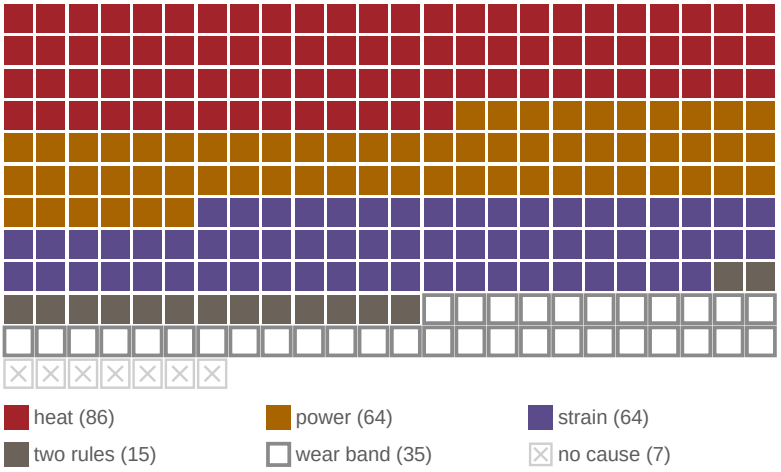

Fig A Every square is one failure. Solid squares are the 229 the rules caught; hollow squares are the 42 the model could not find. Colour shows which rule caught it. Tool wear is not a rule — those failures happen only some of the time.

Why features, not models

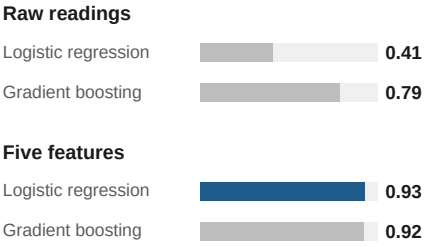

Fig B The same two models, before and after feature engineering. Raw readings take logistic regression to 0.41; the five features take it to 0.93. Model choice moves the third decimal, features move the first.

Three hard rules, one soft one

Three rules are exact — every row they flag is a real failure. Tool wear is different: 624 rows sit in the 200–239 minute band and only 16% of them failed, so there is no sharp line to find.

What causes failures, and what the model uses

Feature	What it checks	Really fail
Heat dissipation hdf	Temperature gap (process minus air) under 8.6 K and spindle speed under 1380 rpm — the machine cannot shed its heat. Flags 94 rows.	100%
Power pwf	Torque × speed × 2π/60 outside 3500 to 9000 W. Flags 73 rows.	100%
Overstrain osf	Tool wear × torque over the limit for that type (L 11000, M 12000, H 13000). Flags 77 rows.	100%
Tool wear wear_pos	How far into the 200 to 239 minute band, from 0 to 1. 624 rows sit in it and only 16% fail, so there is no sharp line — only a rising risk.	16%
Product type L is_L	1 for type L, which fails 3.8% of the time against 2.9% for M and 2.2% for H (4,816 rows).	3.8%

Short names	The grey short names above are the column names in the code: **hdf** heat dissipation failure, **pwf** power failure, **osf** overstrain failure, **wear_pos** position in the wear band, **is_L** product type L.
Scores	Guessing at random scores **0.034** PR-AUC, so 0.931 is a large gain. **Precision 99.6%** = at most one healthy row was flagged. **Recall 84.5%** = 42 of the 271 real failures slipped through. F1 balances the two.

How we tested it

Stratified 5-fold cross-validation: the data is cut into five parts, the model trains on four and predicts the fifth, so every row is scored by a model that never saw it. We repeat with 20 random splits, because 271 failures is a small number and one split can be lucky.

- **The features did the work.** Fig B: with them, both model families go above 0.92. Model choice moves the last thousandths.
- **Nothing beat logistic regression.** Fig C: we ran twelve standard families on the same features. The best rival was a neural network at 0.929, against 0.931.
- **Precision and recall are a dial, not a fact.** Fig D: pushing recall from 84.5% to 91.5% catches 19 more failures but costs 146 false alarms. We ship the 0.5 cut-off because F1 is highest there.
- **At most one false alarm.** Fig E: 7,727 of 7,728 healthy rows correctly left alone. That one row is genuinely borderline, not a rounding artefact — see Fig E.
- **The score is not tuned into existence.** Re-choosing the regularisation strength *inside* every fold still gives PR-AUC 0.930. Training on only 10% of the rows gives 0.931, and swapping the halves gives 0.931.
- **We did not round the temperature gap.** The labels follow the raw binary arithmetic the generator used; rounding to exact decimals loses 11 real failures and drops PR-AUC to about 0.90.

What this means in practice

The heat, power and overstrain alerts can be trusted — every row they flagged was a real failure, so a team would not chase false alarms. Tools in the 200 to 239 minute band fail only about 16% of the time, so replace them at the next planned stop rather than halting the line. Fig F shows the three exact rules carrying the largest weights, so an engineer can check the model against the physics.

Limits of these results

Every score here comes from splitting *train.csv* itself, and we took the rule limits from that same file. So the honest claim is narrow: **this model works on machines like this one** — the same L, M and H product types, run the same way. A machine with genuinely different characteristics is a different problem: these cut-offs are physical constants of this design, and a different spindle, tool or duty cycle shifts every one. The five features stay the right idea, but each threshold needs re-fitting on the new machine's own data. On the hidden set — same columns, so we assume the same machine — expect PR-AUC around 0.93 and F1 nearer 0.90 to 0.91. On this data, recall above about 0.97 looks unreachable: 7 failures break no rule at all.

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229 of 271 failures found, at most one false alarm, and the rest look random — close to the ceiling on this data, not a shortfall.

Did anything else beat it?

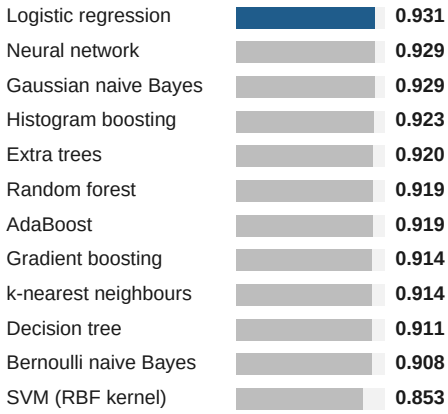

Fig C Twelve model families, same features, same cross-validation. The top three are within 0.003 of each other, which is the point: once the features carry the physics, the classifier barely matters.

The precision / recall dial

cut-off	precision	recall	alarms
0.50	**99.6%**	**84.5%**	1
0.30	**97%**	**86%**	7
0.10	**85%**	**88%**	41
0.03	**63%**	**91.5%**	147

Fig D One model, four cut-offs. Lifting recall 7 percentage points catches 19 more failures and costs 146 extra false alarms.

What it caught and missed

Healthy, correctly left alone **7,727**	False alarm **1**
Failure we missed **42**	Failure we caught **229**

Fig E Out-of-fold at the 0.5 cut-off, averaging 5 splits. Averaging 20 instead takes the false alarm to 0 and F1 to 0.916; one borderline row is all that separates them.

What the model leans on

feature	weight
hdf	**+0.94**
pwf	**+0.82**
osf	**+0.77**
wear_pos	**+0.54**
is_L	**-0.08**

Fig F The fitted weights. The three exact rules carry the most weight; product type barely counts at all.